Jagadeesh V

251901064

Ex.No:16	**NMAP TO DISCOVER LIVE HOSTS USING ARP, ICMP, TCP AND UDP PING SCANS IN TRYHACKME**
DATE:	

1. Aim

To discover live hosts in a network using Nmap by performing ARP scan, ICMP scan, TCP ping scan, and UDP ping scan through a free TryHackMe learning environment.

2. Software Requirements

- Computer or laptop with an internet connection
- Web browser
- Free TryHackMe account
- A free-access lab room with an authorized target
- Nmap terminal in the AttackBox or an authorized lab machine

3. Procedure

Step 1: Create a free TryHackMe account

Open https://tryhackme.com in a web browser. Select Sign Up, register using an available sign-in method, and log in to the account.

Step 2: Select a free room

Visit **https://tryhackme.com/module/nmap** and look for a room or activity related to Nmap and host discovery. Confirm that it is available under the free plan. If the relevant activity requires a subscription, choose another free room that provides a suitable authorized lab target. The learning-path catalogue is available at https://tryhackme.com/paths.

Step 3: Start the lab environment

Open the selected room, read its instructions, and start the target machine. Start the browser-based AttackBox if it is included in the available plan. Note the target IP address shown by the room and open the terminal. If using a local terminal, ensure it can reach the authorized lab target.

Step 4: Verify Nmap installation

Enter the following command in the terminal:

nmap --version

Step 5: Perform ARP scan
Command:

sudo nmap -PR -sn TARGET_IP

Explanation:

- -PR: Uses ARP requests for host discovery.
- -sn: Discovers hosts without performing a port scan.
- TARGET_IP: Replace this placeholder with the IP address of the authorized lab target.

Observation: Record whether Nmap identifies the target as up. ARP discovery generally applies to the same local network segment.

Step 6: Perform ICMP scan
Command:

sudo nmap -PE -sn TARGET_IP

Explanation: -PE sends ICMP Echo Request probes. The -sn option performs host discovery without port scanning.

Observation: Record whether Nmap reports "Host is up". A firewall may block ICMP requests even when a host is active.

Step 7: Perform TCP ping scan
Command:

sudo nmap -PS80,443 -sn TARGET_IP

Explanation: -PS sends TCP SYN probes to ports 80 and 443. The -sn option disables port scanning.

Observation: Record whether Nmap detects the target as reachable. The result depends on target configuration and firewall rules.

Step 8: Perform UDP ping scan
Command:

sudo nmap -PU53,123 -sn TARGET_IP

Explanation: -PU sends UDP host-discovery probes to ports 53 and 123. The -sn option performs host discovery without port scanning.

Observation: Record the actual output. UDP probes may receive no response even when the target is active.

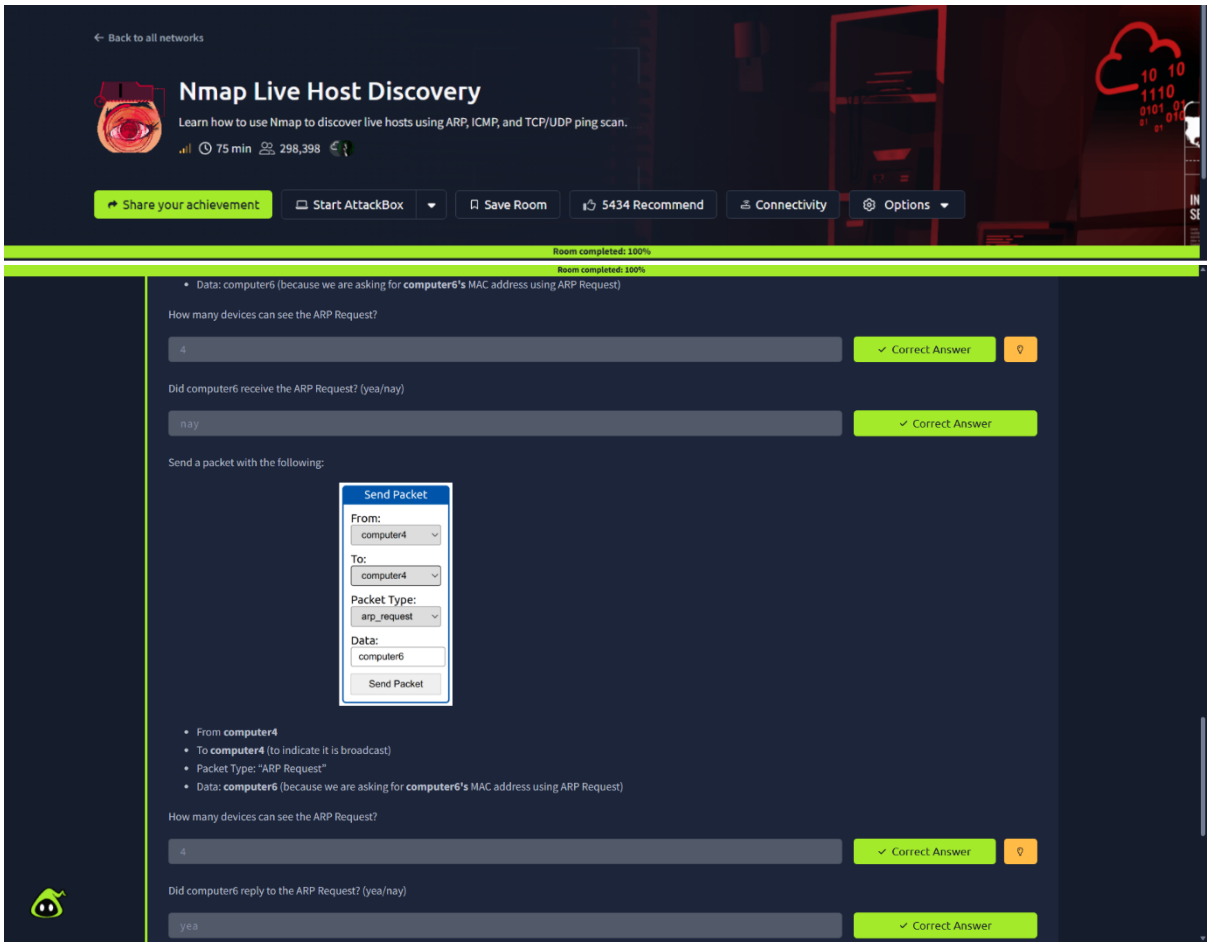
4. Observation Table

S. No.	Scan Technique	Nmap Option	Observed Result
1	ARP scan	-PR	_____
2	ICMP scan	-PE	_____
3	TCP SYN ping	-PS	_____

4	UDP ping	-PU	

Expected observation table

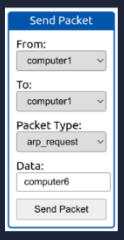
S. No.	Scan technique	Command option	Expected observation
1	ARP scan	`-PR`	Host detected on the same local network
2	ICMP scan	`-PE`	Host detected if ICMP response is received
3	TCP SYN ping	`-PS`	Host detected if a suitable TCP response is received
4	UDP ping	`-PU`	Host detected if a suitable response is received



Room completed: 100%

Answer the questions below

Send a packet with the following:



- From **computer1**
- To **computer1** (to indicate it is broadcast)
- Packet Type: "ARP Request"
- Data: computer6 (because we are asking for **computer6's** MAC address using ARP Request)

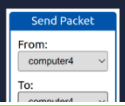
How many devices can see the ARP Request?

4 ✓ Correct Answer

Did computer6 receive the ARP Request? (yea/nay)

nay ✓ Correct Answer

Send a packet with the following:



Room completed: 100%

Task 3 Understanding Hosts Discovery through TCP/IP Layer

Task 4 Enumerating Targets

We mentioned the different techniques we can use for scanning in Task 1. Before we explain each in detail and test it against a live target, we need to specify the targets we want to scan. Generally speaking, you can provide a list, a range, or a subnet. Examples of target specification are:

- list: `MACHINE_IP scanme.nmap.org example.com` will scan 3 IP addresses.
- range: `10.11.12.15-20` will scan 6 IP addresses: `10.11.12.15`, `10.11.12.16`,... and `10.11.12.20`.
- subnet: `MACHINE_IP/30` will scan 4 IP addresses.

You can also provide a file as input for your list of targets, `nmap -iL list_of_hosts.txt`.

If you want to check the list of hosts that Nmap will scan, you can use `nmap -sL TARGETS`. This option gives you a detailed list of hosts that Nmap will scan without actually scanning them; however, Nmap will attempt reverse-DNS resolution on all targets to obtain their names. Names might reveal various information to the pentester. (If you don't want Nmap to use the DNS server, you can add `-n`.)

Connecting to the Network

In this room, you will work with a small network consisting of one Windows host and one Ubuntu host to practice identifying live systems on a network.

When you open the room, you will see a network diagram showing the available hosts. To begin, click the `Start` button to deploy the network. Allow up to 2 minutes for all machines to initialise properly. Once the network has finished loading, launch the AttackBox by clicking the `Start AttackBox` button. When the AttackBox opens in split view, open a Terminal and run `Nmap` to perform host discovery, then answer the questions in this room.

Answer the questions below

What is the first IP address Nmap would scan if you provided 10.10.12.13/29 as your target?

10.10.12.8 ✓ Correct Answer

How many IP addresses will Nmap scan if you provide the following range: 10.10.0-255.101-125?

6400 ✓ Correct Answer

Room completed: 100%

Answer the questions below

Send a packet with the following:
From **computer1**
To **computer3**
Packet Type: "**Ping Request**"
What type of packet did **computer1** send before the ping?

ARP Request — ✓ Correct Answer

What type of packet did **computer1** receive before it was able to send the ping?

ARP Response — ✓ Correct Answer

How many computers responded to the ping request?

1 — ✓ Correct Answer

Send a packet with the following:
From **computer2**
To **computer5**
Packet Type: "**Ping Request**"
What is the name of the first device that responded to the first ARP Request?

router — ✓ Correct Answer

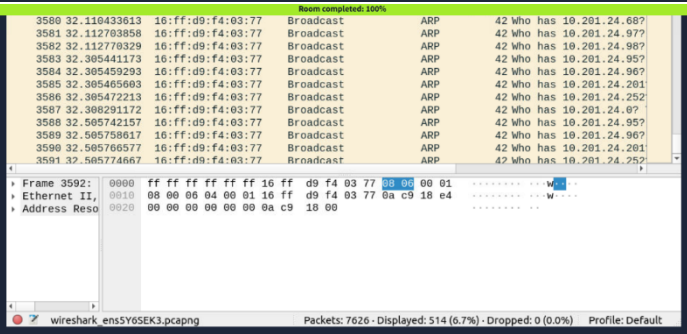
What is the name of the first device that responded to the second ARP Request?

computer5 — ✓ Correct Answer

Send another Ping Request. Did it require new ARP Requests? (yea/nay)

nay — ✓ Correct Answer

Room completed: 100%



The source MAC address belongs to the AttackBox, while the destination MAC address is the broadcast address. This is expected, as the target host's MAC address is not yet known. However, the target IP address being queried is visible in the Info column. As shown in the figure, ARP requests are sent sequentially to each IP address on the subnet, starting from `10.201.24.1`.

When a host with the queried IP address is online, it responds with an ARP reply containing its MAC address. Receiving this reply confirms that the host is active and reachable on the network.

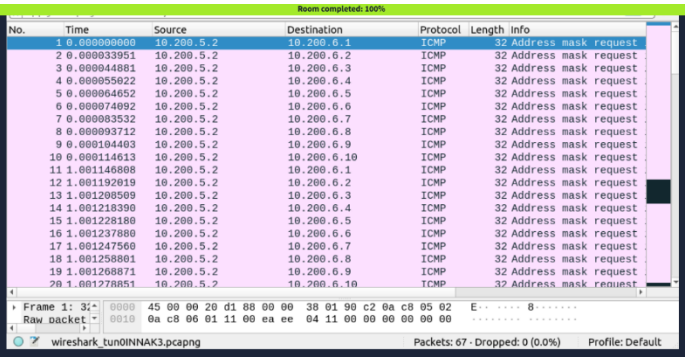
ARP scanning is particularly useful during post-exploitation and internal network enumeration. Once an attacker gains access to a system within a network, ARP scans can be used to quickly and reliably identify other live hosts on the same local segment. Because ARP operates at Layer 2 and is often not filtered by firewalls, it is an effective tool for red-team operations and internal penetration testing.

Answer the questions below

How many hosts are found alive after scanning the CONNECTION_IP/24?

1 — ✓ Correct Answer

Room completed: 100%



Answer the questions below

What is the option required to tell Nmap to use ICMP Timestamp to discover live hosts?

-PP ✓ Correct Answer

What is the option required to tell Nmap to use ICMP Address Mask to discover live hosts?

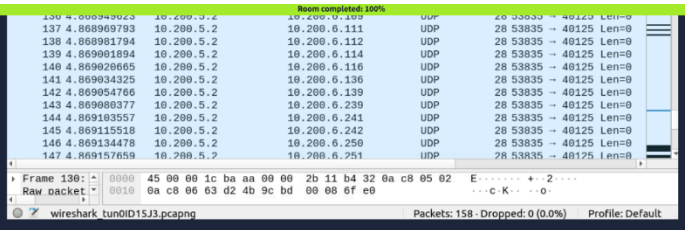
-PM ✓ Correct Answer

What is the option required to tell Nmap to use ICMP Echo to discover live hosts?

-PE ✓ Correct Answer

Task 7 Nmap Host Discovery Using TCP and UDP

Room completed: 100%



Masscan

On a side note, Masscan uses a similar approach to discover the available systems. However, to complete its network scan quickly, Masscan is quite aggressive in the rate at which it generates packets. The syntax is quite similar: `-p` can be followed by a port number, list, or range. Consider the following examples:

- `masscan 10.200.6.0/24 -p443`
- `masscan 10.200.6.0/24 -p80,443`
- `masscan 10.200.6.0/24 -p22-25`

Masscan is not installed on the `AttackBox`; however, it can be installed using `apt install masscan`.

Answer the questions below

Which TCP ping scan does not require a privileged account?

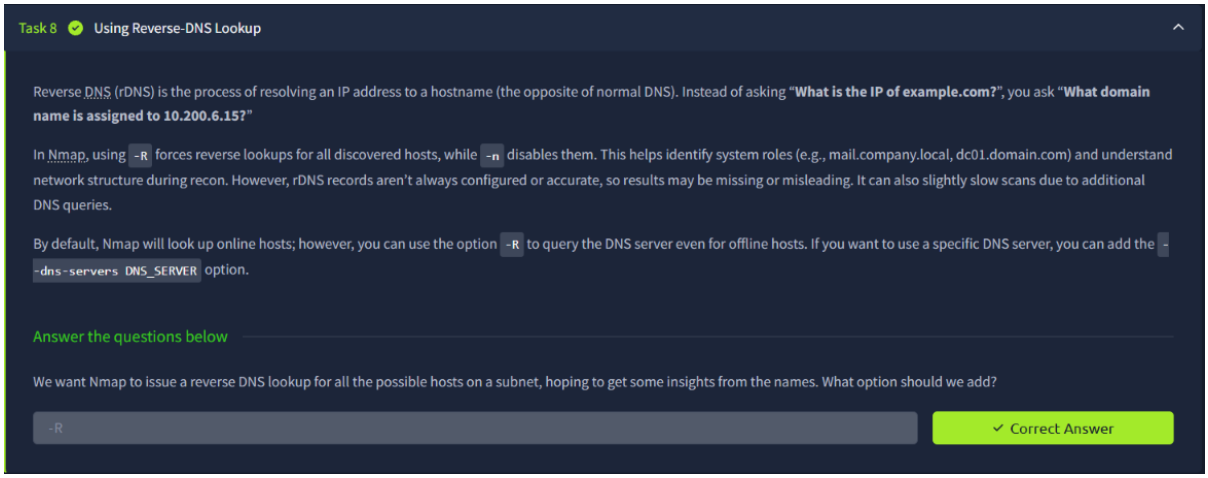
TCP SYN Ping ✓ Correct Answer

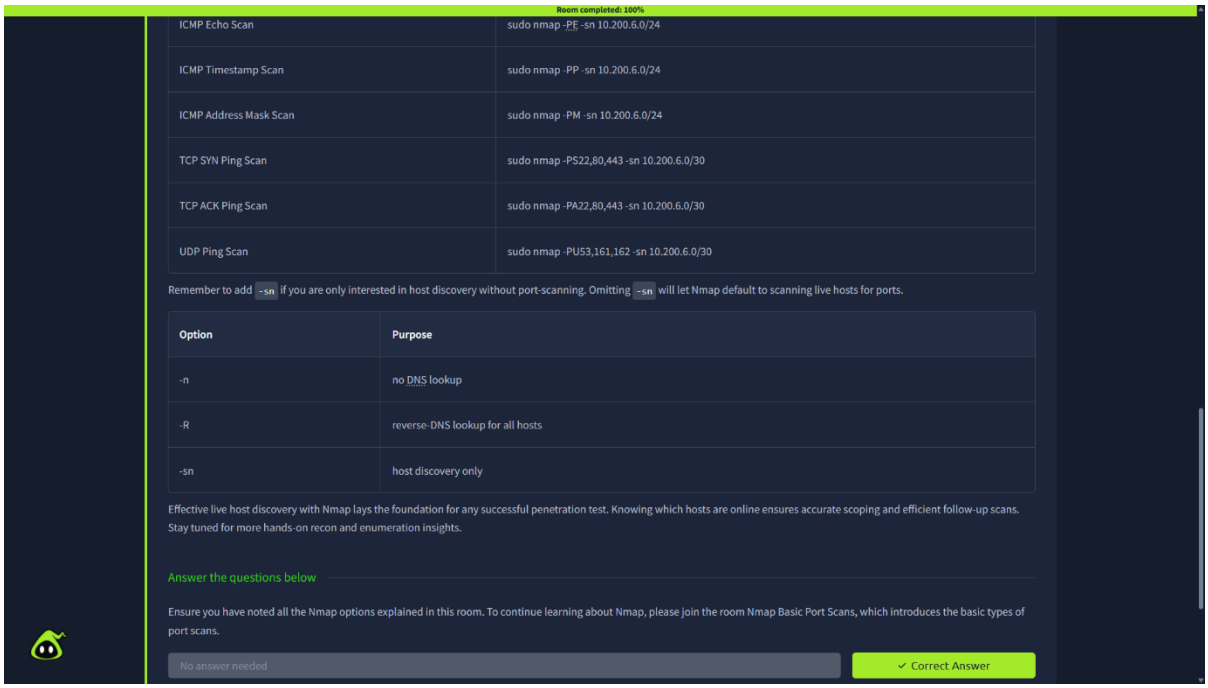
Which TCP ping scan requires a privileged account?

TCP ACK Ping ✓ Correct Answer

What option do you need to add to Nmap to run a TCP SYN ping scan on the telnet port?

-PS23 ✓ Correct Answer





5. Result

Live host discovery was studied using Nmap in the TryHackMe learning environment. ARP, ICMP, TCP, and UDP probes were used to examine host reachability, and the observations were recorded.

6. Conclusion

Nmap provides multiple techniques for discovering active hosts. The effectiveness of each technique depends on network topology, target configuration, and firewall rules.